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MUTING TOKENS: EXPLORING TOKEN SPARSIFICATION FOR EFFICIENT AUDIO TRANSFORMERS

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ABSTRACT

Transformer-based models have shown strong performance across diverse tasks, but their reliance on self-attention mechanisms often leads to high computational costs. Token sparsification, which reduces the number of tokens processed in transformer layers, offers a promising solution to maintain performance while improving efficiency. However, although widely studied in the vision and language domains, token sparsification remains underexplored in audio, with existing work often limited to a single method, model size, or dataset. This work presents a comprehensive evaluation of token sparsification techniques for audio transformers. We assess a range of token pruning and merging strategies across multiple datasets and model sizes, under both post-training and fine-tuning settings. Furthermore, we introduce a novel sparsification approach that combines attention scores with spectral cues derived from the frequency content of token embeddings. Our experiments show that frequency-aware approaches, particularly those based on spectral flatness and low-frequency energy, consistently outperform attention-only methods. We also find that sparsification based on spectrogram patch intensity is especially effective for longer audio inputs. These results highlight the value of incorporating frequency-domain and input-specific information into sparsification strategies to enhance performance in audio tasks.

1 INTRODUCTION

Transformer-based models have demonstrated the ability to achieve state-of-the-art performance across various domains such as Natural Language Processing (NLP) (Devlin et al., 2019; Vaswani et al., 2017) and Computer Vision (Khan et al., 2022; Radford et al., 2021). However, the self-attention mechanism can be computationally intensive, as its complexity scales quadratically with the number of input tokens (Bolya et al., 2023; Fayyaz et al., 2022). To address this challenge, recent work has explored compression techniques such as pruning (Chavan et al., 2022; Song et al., 2022), quantization (Li et al., 2022b; Lin et al., 2023), and knowledge distillation (KD) (Rao et al., 2021b; Ren et al., 2021), in an effort to reduce model size and inference latency while maintaining task performance.

Recently, token sparsification, a transformer-specific optimization method that discards, merges, or refines input tokens, has emerged as a promising alternative to reduce the

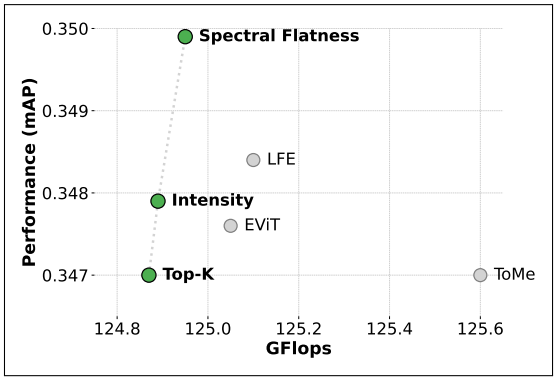


Figure 1. Post-training sparsification results on the AST-Base (Gong et al., 2021) model, evaluated on AudioSet-20K (Gemmeke et al., 2017) at a 50% keep rate. Methods that include frequency-domain cues, such as the proposed **Spectral Flatness (SF)** and the adapted **Low-Frequency Energy (LFE) (Wang et al., 2022b)**, outperform others in terms of mean average precision (mAP), a standard metric for multi-label classification. The baseline (without sparsification) achieves an average mAP of 0.3572. The Pareto front is highlighted to illustrate the performance-efficiency trade-off.

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静音令牌：探索令牌稀疏化以实现高效音频变换器

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摘要

基于Transformer的模型在各种任务中表现出强大的性能，但它们对自注意力机制的依赖通常导致高计算成本。Token稀疏化通过减少Transformer层中处理的tokens数量，提供了一种在保持性能的同时提高效率的有前景的解决方案。然而，尽管在视觉和语言领域得到了广泛研究，token稀疏化在音频领域仍然研究不足，现有工作通常仅限于单一方法、模型大小或数据集。本工作对音频Transformer的token稀疏化技术进行了全面评估。我们在多种数据集和模型规模下评估了一系列token剪枝和合并策略，涵盖后训练和微调两种设置。此外，我们提出了一种新颖的稀疏化方法，将注意力分数与由token嵌入频率内容导出的频谱特征相结合。我们的实验表明，频率感知的app

1 引言

基于Transformer的模型已展示出在各个领域中实现最先进性能的能力，例如自然语言处理（NLP）（Devlin 等, 2019；Vaswani 等, 2017）和计算机视觉（Khan 等, 2022；Radford 等, 2021）。然而，自注意力机制可能计算密集，因为其复杂度随输入令牌数量呈二次增长（Bolya 等, 2023；Fayyaz 等, 2022）。为了解决这一挑战，近期的研究探索了压缩技术，例如剪枝（Chavan 等, 2022；Song 等, 2022）、量化（Li 等, 2022b；Lin 等, 2023）和知识蒸馏（KD）（Rao 等, 2021b；Ren 等, 2021），以努力在保持任务性能的同时减少模型大小和推理延迟。

最近，token稀疏化，这是一种针对变换器的优化方法，它丢弃、合并或改进输入token，已经成为一种有前景的替代方案以减少

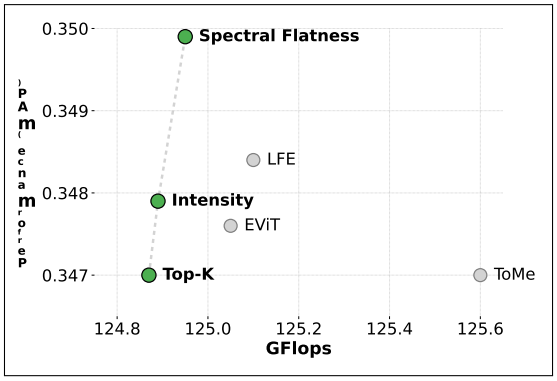


图 1. 在 AST-Base（Gong 等，2021）模型上的训练后稀疏化结果，在 AudioSet-20K（Gemmeke 等，2017）上以 50% 保留率进行评估。包含频域线索的方法，如所提出的谱平坦度（Spectral Flatness, SF）和改进的低频能量（Low-Frequency Energy, LFE）（Wang 等，2022b），在平均精度（mean average precision, mAP）方面优于其他方法，mAP 是多标签分类的标准指标。基线（未稀疏化）实现的平均 mAP 为 0.3572。帕累托前沿被突出显示，以说明性能与效率的权衡。

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computational cost of self-attention. For example, the Expediting Vision Transformers (EViT) method achieves up to 50% faster inference on ImageNet (Deng et al., 2009), with only a 1.3 percentage point drop in accuracy when retaining 50% of the tokens, referred to as the *keep rate* (Liang et al., 2022a).

A growing body of work has explored token sparsification techniques across various tasks, particularly in image classification (Bonnaerens & Dambre, 2023; Chen et al., 2022b; Huang et al., 2024; Kong et al., 2022; Liang et al., 2022a; Wu et al., 2023), video classification (Ryoo et al., 2022; Wu et al., 2024), and even large language models (Cao et al., 2023; Chen et al., 2024; Jiang et al., 2023; Liu et al., 2023). Although Vision Transformers (ViT) have demonstrated notable success in audio tasks by using spectrograms as input representations, the application of token sparsification in this domain remains largely underexplored (Chen et al., 2022a; Gong et al., 2021; 2022a;b; Huang et al., 2022). This gap is particularly important because spectrogram inputs yield more tokens as the input duration increases, which can make self-attention increasingly expensive for longer audio clips (Gong et al., 2021). Addressing this challenge could not only help advance efficient transformers' research, but also enable practical, large-scale deployment of audio-based systems, from everyday voice assistants to real-time transcription services at the edge.

Recent work (Lee & Lee, 2025) has begun to explore the effectiveness of token sparsification in audio tasks. However, while showing promising results, this study presents considerable limitations. First, it restricts its scope to only two sparsification methods, Top-K pruning (Haurum et al., 2022a) and Intensity pruning (Lee & Lee, 2025), overlooking other promising approaches such as token merging (Bolya et al., 2023). Second, their analysis is restricted to only the ViT-Base model (Dosovitskiy et al., 2021) limiting the generalizability of their conclusions to a single model size. Third, the study focuses solely on applying sparsification during fine-tuning, whereas many real-world deployment scenarios favor post-training sparsification methods, mainly due to the widespread availability of pretrained models, and the training costs associated with larger datasets (Chen et al., 2025; Ilhan et al., 2024). Finally, it reports suboptimal performance of the intensity pruning method at low keep rates, without investigating the potential influence of input-dependent factors such as the audio duration.

In this work, we systematically evaluate a broad range of token sparsification methods for audio transformers, examining their performance across multiple model sizes and datasets, under two key settings: fine-tuning and post-training sparsification. To ensure a fair comparison, we conduct all experiments within a unified training and inference pipeline, establishing a consistent benchmark for evaluating sparsification techniques in this domain. We also revisit the intensity pruning method and show that its effectiveness improves when combined with attention scores. Furthermore, we propose a novel sparsification approach that integrates attention and frequency-based cues, drawing on insights from audio signal processing. As illustrated in Fig. 1, methods that incorporate frequency-domain information consistently outperform existing sparsification approaches across our experimental setup. Our contributions can be summarized as:

- We conduct a comprehensive evaluation of token sparsification methods for audio transformers, including token pruning and merging approaches under a unified evaluation setup.
- We demonstrate the effectiveness of intensity-based pruning, showing that when combined with attention scores, it can outperform pure attention methods for audio inputs longer than 2 seconds across our experimental setup.
- We introduce a novel sparsification method based on spectral flatness, and show that it can consistently improve performance across a wide range of model sizes and datasets.

2 Related Work

2.1 Token Sparsification: Overview

Token sparsification reduces the number of tokens processed within transformer layers, with existing methods broadly falling into two main categories (Haurum et al., 2022a). ***Token pruning*** methods (Dou et al., 2023; Fayyaz et al., 2022; Kong et al., 2022; Li et al., 2022a; Liang et al., 2022b; Long et al., 2022; Pan et al., 2021; Tang et al., 2022; Xu et al., 2022b; Yin et al., 2022) reduce the number of tokens by completely removing tokens, at the risk of discarding useful information. In contrast, ***token merging*** methods (Bolya et al., 2023; Haurum et al., 2022b; Marin et al., 2021; Renggli et al., 2022; Ryoo et al., 2021; Wu et al., 2021; Xu et al., 2022a; Zeng et al., 2022; Zong et al., 2022) combine multiple tokens into a single token or cluster, but may introduce noise through averaging.

Our study includes representative methods from both sparsification categories to ensure diversity in evaluation, addressing the lack of systematic assessment in audio tasks.

2.2 Token Sparsification Methods for Audio

Recent efforts to apply token sparsification to audio tasks have primarily adapted techniques originally developed for vision models. Applied on a post-training manner, Token Merging (ToMe) (Bolya et al., 2023) merges similar tokens

自注意力的计算成本。例如，加速视觉变换器（EViT）方法在 ImageNet（Deng 等，2009）上的推理速度可提高最多 50%，在保留 50% 的令牌时准确率仅下降 1.3 个百分点，这被称为 *keep rate* (Liang 等，2022a)。

越来越多的研究工作已经探讨了在各种任务中使用的令牌稀疏化技术，特别是在图像分类（Bonnaerens & Dambre, 2023; Chen et al., 2022b; Huang et al., 2024; Kong et al., 2022; Liang et al., 2022a; Wu et al., 2023）、视频分类（Ryoo et al., 2022; Wu et al., 2024）甚至大型语言模型（Cao et al., 2023; Chen et al., 2024; Jiang et al., 2023; Liu et al., 2023）中。尽管视觉变换器（ViT）通过使用频谱图作为输入表示在音频任务中取得了显著成功，但在该领域中令牌稀疏化的应用仍然很少被探讨（Chen et al., 2022a; Gong et al., 2021; 2022a;b; Huang et al., 2022）。这一空白尤为重要，因为随着输入时长的增加，频谱图输入会产生更多的令牌，这可能使长音频片段的自注意力计算变得越来越昂贵（Gong et al., 2021）。解决这一挑战不仅有助于推动高效变换器的发展

最近的研究（Lee & Lee, 2025）开始探索令牌稀疏化在音频任务中的有效性。然而，尽管显示出有前景的结果，这项研究仍存在相当大的局限性。首先，它的研究范围仅限于两种稀疏化方法：Top-K剪枝（Haurum et al., 2022a）和强度剪枝（Lee & Lee, 2025），忽略了其他有前景的方法，如令牌合并（Bolya et al., 2023）。其次，他们的分析仅限于ViT-Base模型（Dosovitskiy et al., 2021），限制了其结论在不同模型规模上的普适性。第三，该研究仅关注在微调期间应用稀疏化，而许多现实世界的部署场景更偏向于训练后稀疏化方法，主要是由于预训练模型的广泛可用性以及使用更大数据集的训练成本（Chen et al., 2025; Ilhan et al., 2024）。最后，它报告了强度剪枝方法的次优性能。

在本工作中，我们系统地评估了适用于音频变换器的广泛令牌稀疏化方法，考察它们在多种模型规模和数据集上的性能，在两个关键设置下：微调和训练后稀疏化。为了确保公平比较，我们在统一的训练和推理流程中进行所有实验，建立了评估的统一基准。

该领域的稀疏化技术。我们还重新研究了强度剪枝方法，并展示了当与注意力分数结合时，其效果会有所提升。此外，我们提出了一种新颖的稀疏化方法，该方法结合了注意力和基于频率的线索，借鉴了音频信号处理的见解。如图1所示，融合频域信息的方法在我们的实验设置中始终优于现有的稀疏化方法。我们的贡献可以总结如下：

- 我们在统一的评估设置下，对音频变换器的令牌稀疏化方法进行了全面评估，包括令牌剪枝和合并方法。
- 我们展示了基于强度的剪枝的有效性，表明当与注意力分数结合时，它可以在我们的实验设置中，对于超过2秒的音频输入，优于纯注意力方法。
- 我们提出了一种基于谱平坦度的新型稀疏化方法，并表明它可以在各种模型规模和数据集上持续提高性能。

2 相关工作

2.1 令牌稀疏化：概述

令牌稀疏化减少了在变压器层中处理的令牌数量，现有方法大致可分为两大类（Haurum 等，2022a）。***Token pruning*** 方法（Dou 等，2023；Fayyaz 等，2022；Kong 等，2022；Li 等，2022a；Liang 等，2022b；Long 等，2022；Pan 等，2021；Tang 等，2022；Xu 等，2022b；Yin 等，2022）通过完全移除令牌来减少令牌数量，但有丢失有用信息的风险。相反，***token merging*** 方法（Bolya 等，2023；Haurum 等，2022b；Marin 等，2021；Renggli 等，2022；Ryoo 等，2021；Wu 等，2021；Xu 等，2022a；Zeng 等，2022；Zong 等，2022）将多个令牌合并为单个令牌或簇，但可能通过平均化引入噪声。

我们的研究包括来自两种稀疏化类别的代表性方法，以确保评估的多样性，并解决音频任务中缺乏系统性评估的问题。

2.2 音频的令牌稀疏化方法

最近将令牌稀疏化应用于音频任务的努力主要是改编了最初为视觉模型开发的技术。在后训练方式下应用时，令牌合并（ToMe）（Bolya 等，2023）会合并相似的令牌

to reduce the input size, achieving substantial throughput gains with merely a 0.4 percentage point drop in mAP on AudioSet-2M (Gemmeke et al., 2017). However, its evaluation is limited to a single dataset and a masked autoencoding (MAE) framework (He et al., 2021; Huang et al., 2022), which is inherently designed to operate on sparse inputs. This may reduce the model's sensitivity to sparsification, making it unclear how ToMe performs in settings that rely on full token visibility. Without evaluation on non-MAE models, its general applicability remains uncertain, an aspect we address in this work.

A different line of work explores sparsification in audio transformers during training. PaSST (Gong et al., 2022a) employs "patchout" which randomly drops a subset of spectrogram patches to reduce computation and act as a regularizer. While effective, these approaches require training with patchout, limiting their practicality. In contrast, post-training sparsification offers a broadly applicable and cost-efficient alternative, especially when retraining is costly or infeasible.

2.3 Studies on Token Sparsification for Audio

Beyond these early efforts, recent work has begun to explore sparsification methods specifically for audio. As mentioned in Sec. 1, the authors in (Lee & Lee, 2025) investigate Top-K pruning and Intensity pruning. While this represents a promising direction, the study remains narrow in scope, focusing on just two methods, a single ViT-Base configuration, and exclusively on fine-tuning scenarios. This highlights the need for broader studies that evaluate different sparsification approaches, model sizes, and across both fine-tuning and post-training contexts, a gap our work seeks to address.

3 METHODOLOGY

In this study, we pursue two complementary directions. First, we evaluate established token sparsification methods developed for vision, in audio tasks, providing the first systematic comparison of their behavior across audio benchmarks. Second, we propose lightweight, task-specific token importance estimation strategies that incorporate audio-relevant cues such as spectrogram intensity and frequency-domain features.

3.1 Token Sparsification Methods

We evaluate three sparsification methods: Top-K (Haurum et al., 2022a), Expediting Vision Transformers (EViT) (Liang et al., 2022a) and ToMe (Bolya et al., 2023). These methods were selected based on their strong performance in vision tasks (Haurum et al., 2022a) and their representation of the two main paradigms in token sparsification: *pruning* (Top-K and EViT) and *merging* (ToMe). Note that

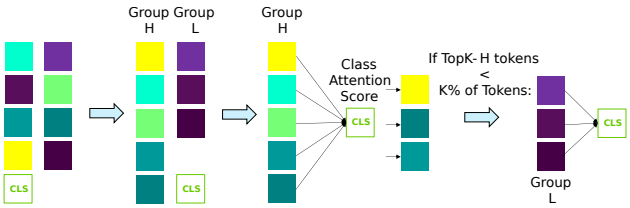


Figure 2. Intensity-guided token pruning. Tokens are divided into high- (H) and low- (L) intensity groups based on an applied threshold. The high-intensity set is then ranked using CLS attention scores, and the Top-K tokens are selected. If fewer than K tokens are available in H, the remaining slots are filled from the low-intensity set.

while some of these methods have been applied to audio in a limited setting, as mentioned in Sec. 2.2, they have not yet been systematically compared or evaluated across benchmarks. **Top-K** is a widely used baseline, where the attention between spatial tokens and the class (CLS) token are used to rank token importance. At each reduction stage s, the K_s most attended tokens are kept. **EViT** extends this approach by introducing a fused token that preserves information from discarded tokens. At each stage s, tokens not in the Top-K set are averaged and weighted by their CLS attention scores. **ToMe** reduces tokens by merging similar token pairs. The token set is split into two partitions, and each token is matched to its most similar counterpart based on cosine similarity. The top matching pairs are then merged by averaging their features.

3.2 Task-specific Token Importance Estimation Strategies

Most sparsification methods rely on internal model signals, such as attention weights or intermediate representations like attention keys. However, in post-training settings where model parameters remain fixed, we hypothesize that lightweight, task-specific heuristics, can more effectively identify informative tokens, leading to improved performance under sparsification. In audio tasks, spectrograms encode energy across time and frequency, but these patterns may not be fully captured by attention alone (Ristea et al., 2022). Prior work highlights the utility of frequency-domain processing in transformers (Lee-Thorp et al., 2021; Patro et al., 2025; Rao et al., 2021a; Tamkin et al., 2020), and that structure may not always be visible in raw spectrograms (Atlas, 2003; Ezzat et al., 2007). Motivated by this, we explore three lightweight heuristics for token importance, one input patch intensity-based, and two frequency-based, that can be combined with many existing methods: 1) intensity, 2) low-frequency energy (Wang et al., 2022b) and 3) spectral flatness.

为了减少输入大小，仅在 AudioSet-2M (Gemmeke 等, 2017) 上 mAP 下降 0.4 个百分点的情况下，实现了显著的吞吐量提升。然而，其评估仅限于单一数据集和掩码自动编码 (MAE) 框架 (He 等, 2021; Huang 等, 2022)，而该框架本质上是为稀疏输入设计的。这可能会降低模型对稀疏化的敏感性，使得在依赖完整 token 可见性的环境中 ToMe 的表现尚不清楚。如未对非 MAE 模型进行评估，其通用适用性仍不确定，这是我们在本工作中所要解决的一个方面。

另一类研究在训练过程中探索音频变压器的稀疏化。PaSST（Gong 等，2022a）采用“patchout”，随机丢弃一部分频谱图块以减少计算量并起到正则化作用。虽然有效，但这些方法需要在训练时使用 patchout，限制了其实际应用性。相比之下，训练后稀疏化提供了一种广泛适用且成本高效的替代方案，尤其是在重新训练代价高或不可行时。

2.3 声音的标记稀疏化研究

除了这些早期的努力，最近的工作已经开始专门探索针对音频的稀疏化方法。如第1节所述，（Lee & Lee, 2025）的作者研究了Top-K剪枝和强度剪枝。虽然这代表了一个有前景的方向，但该研究范围仍然有限，仅关注两种方法、一个ViT-Base配置，并且完全针对微调场景。这凸显了需要更广泛的研究，以评估不同的稀疏化方法、模型规模，并涵盖微调和训练后情境，而这正是我们的工作试图解决的空白。

3 方法论

在本研究中，我们追求两个互补的方向。首先，我们评估为视觉开发的已建立的令牌稀疏化方法在音频任务中的表现，这是首次跨音频基准系统地比较它们的行为。其次，我们提出了轻量级的、任务特定的令牌重要性估计策略，这些策略结合了音频相关的线索，如频谱强度和频域特征。

3.1 词元稀疏化方法

我们评估了三种稀疏化方法：Top-K（Haurum 等，2022a）、加速视觉变换器（EViT）（Liang 等，2022a）和 ToMe（Bolya 等，2023）。这些方法的选择基于它们在视觉任务中的出色表现（Haurum 等，2022a）以及它们在令牌稀疏化的两种主要范式中的代表性：*pruning* (Top-K 和 EViT) 以及 *merging* (ToMe)。注意

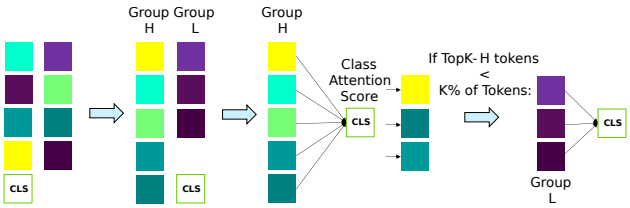


图2. 强度引导的标记剪枝。根据应用的阈值，标记被分为高强度（H）和低强度（L）组。然后使用CLS注意力分数对高强度集合进行排名，并选择排名前K的标记。如果H中可用的标记少于K个，则剩余的空位由低强度集合填充。

虽然如第2.2节所述，这些方法中的一些已在有限的音频场景中应用，但它们尚未在各个基准上进行系统的比较或评估。Top-K 是一个广泛使用的基线方法，其中使用空间 token 与类别（CLS）token 之间的注意力来对 token 的重要性进行排序。在每个降维阶段 s，保留 K_s 个被关注度最高的 token。EViT 通过引入一个融合 token 来扩展这一方法，该融合 token 保留了被舍弃 token 的信息。在每个阶段 s，不在 Top-K 集合中的 token 会被平均处理，并根据其 CLS 注意力分数进行加权。ToMe 通过合并相似的 token 对来减少 token 数量。token 集合被分成两个部分，每个 token 会根据余弦相似度匹配其最相似的对应 token。匹配度最高的 token 对随后通过平均它们的特征进行合并。

3.2 特定任务的令牌重要性评估策略

大多数稀疏化方法依赖于内部模型信号，如注意力权重或中间表示（例如注意力键）。然而，在模型参数保持固定的后训练设置中，我们假设轻量级、任务特定的启发式方法可以更有效地识别信息性令牌，从而在稀疏化下提高性能。在音频任务中，声谱图编码了随时间和频率变化的能量，但这些模式可能无法仅通过注意力完全捕捉（Ristea 等，2022）。先前的研究强调了变换器中频域处理的实用性（Lee-Thorp 等，2021；Patro 等，2025；Rao 等，2021a；Tamkin 等，2020），并且这些结构可能并不总是能在原始声谱图中可见（Atlas，2003；Ezzat 等，2007）。受此启发，我们探索了三种轻量级的令牌重要性启发式方法，一种基于输入补丁强度，两种基于频率，可以与许多现有方法结合：1）强度

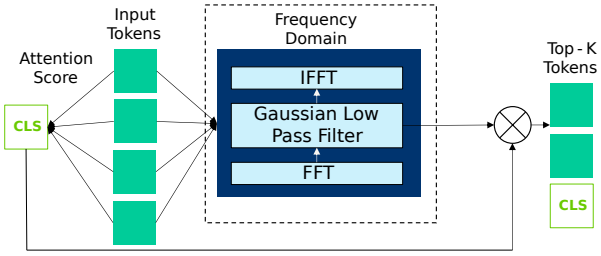


Figure 3. Low-frequency Energy token pruning. Token embeddings are transformed to the frequency domain, where a Gaussian low-pass filter highlights low-frequency structure. The resulting energy scores are fused with attention scores to guide top-K token selection.

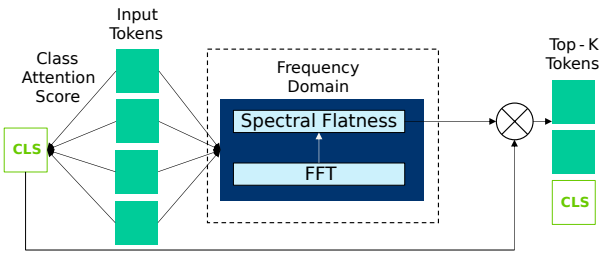


Figure 4. Spectral flatness-guided token pruning. Token embeddings undergo Fast Fourier Transform (FFT) along the embedding dimension to compute spectral flatness, capturing tonal richness. These scores are combined with CLS attention to rank token importance.

Intensity. Following (Lee & Lee, 2025), we compute the mean log-Mel energy of each spectrogram patch as a lightweight salience proxy, based on the hypothesis that high intensity regions reflect stronger signal content. However, using intensity alone can be unreliable, as it can sometimes highlight background noise or non-discriminative events with high energy (Bhattacharjee et al., 2022; Lee & Lee, 2025). We therefore use it as a coarse filter. At each layer, tokens are split into high and low intensity sets based using a dataset-specific threshold. The Top-K tokens are selected by ranking the high-intensity ones via CLS attention scores, with any remainder filled from the low-intensity pool (Fig. 2). This method is primarily intended to test whether incorporating spectrogram intensities improves attention-only sparsification and how it compares to frequency-aware approaches.

Low-Frequency Energy. Inspired by the Low-Frequency Components (LFC) model for vision transformers, we adopt their idea of using low-frequency energy (LFE) for token sparsification (Wang et al., 2022b). In their work, low-pass filtering of token embeddings was shown to preserve task-relevant content more effectively, aligning with the observation that transformers tend to favor low-frequency features (Dosovitskiy et al., 2021; Wang et al., 2022a). We extend this idea to audio by computing low-frequency energy within each token embedding, after patch extraction and projection. This provides a local measure of how much of the token's representation is concentrated in low-frequency components. This score is combined with the token's attention weight, allowing the model to prioritize tokens that are both attended and rich by low-frequency energy, as illustrated in Fig. 3.

Spectral Flatness. Spectral Flatness (SF) is a widely used measure in audio signal processing for quantifying the degree to which a signal resembles noise or contains structured, tonal components (Jayant & Noll, 1993; Seago et al., 2008). It is computed as the ratio of the geometric mean to the arithmetic mean of a signal's magnitude spectrum, with lower values indicating pronounced spectral peaks typical of harmonic or speech-like content, and higher values corresponding to flatter, noise-like distributions (Boashash et al., 2015; Madhu, 2009). Motivated by these properties, we extend the use of SF to transformer token embeddings by computing it for each token, as illustrated in Fig. 4. We treat each embedding as a learned signal, using SF as a proxy for its internal organization, and hypothesize that lower flatness values indicate more structured and content-rich tokens. An illustrative example of this phenomenon is shown in Fig. 5, where two spectrogram patches yield token embeddings with contrasting spectral profiles. The token's flatness score is combined with its attention to the CLS token, prioritizing tokens that are both spectrally structured and highly attended.

3.3 Evaluation Strategy

We aim to systematically evaluate token sparsification methods (Sec. 3.1) across multiple audio benchmarks, while also investigating the role of input- and frequency-aware token importance estimation strategies (Sec. 3.2). To understand both the practical and performance implications of sparsification, we evaluate all methods under two settings:

- **Post-training Sparsification.** Sparsification is applied only during inference, using already finetuned models.
- **Sparsified Fine-tuning.** Sparsification is applied during fine-tuning, starting from ImageNet-pretrained weights and adapting to the target audio dataset.

Using these two settings, we aim to understand the benefits and tradeoffs of sparsified fine-tuning over post-training sparsification, especially at lower keep rates.

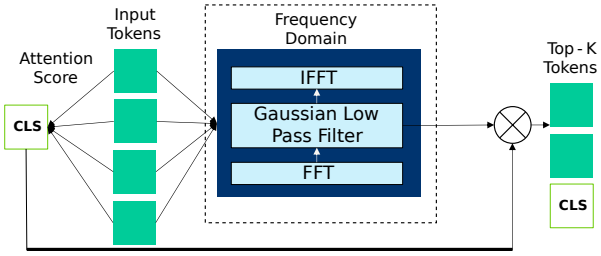


图 3. 低频能量令牌剪枝。令牌嵌入被转换到频域，其中高斯低通滤波器突出低频结构。生成的能量分数与注意力分数融合，以指导前 K 个令牌的选择。

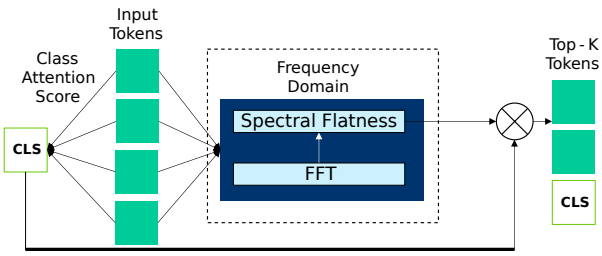


图4. 基于谱平坦度的Token剪枝。Token嵌入沿嵌入维度进行快速傅里叶变换（FFT）以计算谱平坦度，从而捕捉音色丰富性。这些分数与CLS注意力结合，用于排序Token的重要性。

强度。根据（Lee & Lee, 2025），我们计算每个频谱图块的平均对数梅尔能量，作为一种轻量级的显著性代理，基于假设高强度区域反映了更强的信号内容。然而，仅使用强度可能不可靠，因为它有时会突出背景噪声或高能量的非判别性事件（Bhattacharjee 等, 2022；Lee & Lee, 2025）。因此，我们将其用作粗略筛选。在每一层中，基于数据集特定的阈值，将令牌分为高强度集和低强度集。通过 CLS 注意力分数对高强度令牌进行排名，从中选择 Top-K 令牌，剩余部分从低强度池中补充（图 2）。该方法主要旨在测试引入频谱图强度是否能改善仅注意力的稀疏化，以及它与频率感知方法的比较效果。

低频能量。受到视觉变换器中低频分量（LFC）模型的启发，我们采用了他们使用低频能量（LFE）进行标记稀疏化的想法（Wang 等，2022b）。在他们的工作中，对标记嵌入进行低通滤波显示能够更有效地保留任务相关内容，这与观察结果一致，即变换器倾向于偏好低频特征（Dosovitskiy 等，2021；Wang 等，2022a）。我们将这一思想扩展到音频，通过在提取补丁并进行投影后计算每个标记嵌入内的低频能量。这提供了一种局部度量，反映标记表征中有多少集中在低频分量。然后，将该分数与标记的注意力权重结合，使模型能够优先考虑既被关注又富含低频能量的标记，如图3所示。

谱平坦度。谱平坦度（SF）是音频信号处理中广泛使用的度量，用于量化一个信号与噪声的相似程度或包含结构的程度

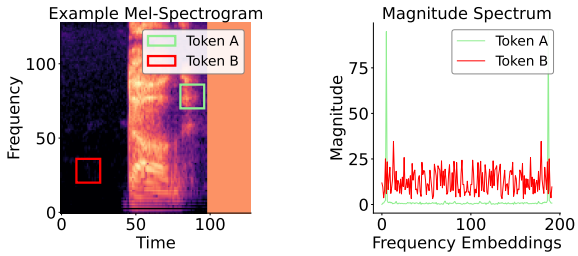
tured，音调成分（Jayant & Noll, 1993；Seago 等，2008）。它是通过信号幅度谱的几何平均数与算术平均数的比值来计算的，较低的值表示典型的谐波或类似语音内容的明显谱峰，而较高的值对应更平坦、类似噪声的分布（Boashash 等，2015；Madhu, 2009）。受这些特性的启发，我们将 SF 的使用扩展到 transformer 令牌嵌入，通过对每个令牌计算它，如图 4 所示。我们将每个嵌入视为一个学习到的信号，使用 SF 作为其内部组织的代理，并假设较低的平坦度值表示更有结构性和内容丰富的令牌。该现象的一个示例如图 5 所示，其中两个频谱图片段产生具有对比谱特征的令牌嵌入。令牌的平坦度分数与其对 CLS 令牌的关注相结合，优先考虑那些在谱上具有结构性的令牌

3.3 评估策略

我们旨在系统地评估多种音频基准中的令牌稀疏化方法（第 3.1 节），同时研究输入感知和频率感知的令牌重要性估计策略的作用（第 3.2 节）。为了理解稀疏化的实际和性能影响，我们在两种设置下评估所有方法：

- 训练后稀疏化。稀疏化仅在推理期间应用，使用已微调的模型。
- 稀疏化微调。在微调期间应用稀疏化，从ImageNet预训练权重开始，并适应目标音频数据集。

使用这两种设置，我们旨在了解稀疏化微调相对于训练后稀疏化的优势和权衡，尤其是在较低保留率时。



(a) Example Mel spectrogram (Warden, 2018).

(b) Corresponding magnitude spectra.

Figure 5. Spectral flatness at the token level. (a) A spectrogram with two highlighted tokens. (b) FFT of corresponding token embeddings shows Token A has tonal peaks, while Token B has a flatter, noise-like spectrum - making Token A potentially more informative.

4 Experimental Setup

Datasets. For this analysis, three well-established datasets are selected, each covering different audio sample lengths and task complexities. **Speech Commands V2 (SPC)** (Warden, 2018) comprises 105,829 **1-second** recordings of 35 spoken commands. The dataset is divided into 84,843 training samples, 9,981 validation samples, and 11,005 test samples. **Environment Sound Classification (ESC50)** (Piczak, 2015) consists of 2,000 **5-second** environmental sound recordings categorized into 50 classes. Its 5-fold cross-validation setup includes 1,600 training samples per fold, ensuring a reliable evaluation framework. **AudioSet (AS-20K)** (Gemmeke et al., 2017) is a large-scale multi-class dataset containing **10-second** audio clips sourced from YouTube, labeled across 527 categories. The balanced training set consists of around 22,000 samples, while the evaluation set includes around 20,000 samples.

Backbone. We adopt the AST (Audio Spectrogram Transformer) (Gong et al., 2021) architecture as our backbone for all experiments. The sparsification methods are integrated into the AST model pretrained on ImageNet (Deng et al., 2009), using the standard DeiT-based (Touvron et al., 2020) configurations: Base (B), Small (S), and Tiny (T), with patch sizes of 16×16 and an overlap of 6 in both time and frequency, consistent with the default AST recipe.

Sparsification Allocation. As in previous work, the token sparsification module is placed between the attention and multi-layer perceptron (MLP) branches of each transformer block of the AST (Bolya et al., 2023). The most common and effective practice is to apply sparsification at three equally spaced points in the transformer, effectively dividing the network into four sections (Kong et al., 2022; Liang et al., 2022b; Long et al., 2022). This is the configuration adopted in our experiments. At each sparsification stage, a fixed keep rate $r \in \{0.25, 0.50, 0.70, 0.90\}$ is applied uniformly across all sparsified layers.

Implementation Details. All training and fine-tuning procedures follow the original AST setup (Gong et al., 2021). Specifically, AS-20K and ESC50 are finetuned for 25 epochs, while SPC is trained for 30 epochs. When token sparsification is enabled, all default masking strategies (e.g., SpecAugment (Park et al., 2019)) in AST are disabled to avoid redundancy, as sparsification inherently performs a form of data masking. All model variants (AST-B, AST-S, AST-T) are evaluated under the post-training sparsification setting to assess performance and efficiency across different model scales. For the sparsified fine-tuning experiments, we use only AST-B, as it is the default model size in the original AST implementation. To quantify the compute cost associated with different sparsification methods, we report the number of Floating Point Operations (FLOPs) per instance using a batch size of 1, measured via the DeepSpeed profiler (Aminabadi et al., 2020).

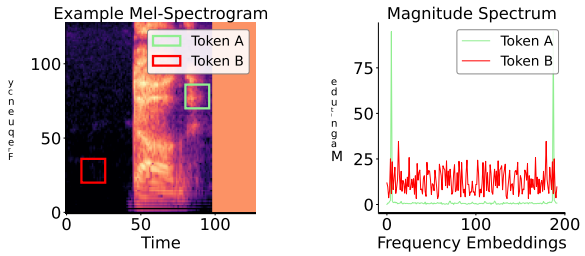
5 Results & Analysis

5.1 Post-Training Sparsification

Performance Across Methods and Model Sizes. Tables 2a, 2b, and 2c report performance across the AST-B, AST-S, and AST-T models, respectively. Results are shown for all datasets (SPC, ESC50, AS-20K) and across token keep rates from 90% down to 25%. We report the average of three runs.

From these results, we find that methods using token frequency information, particularly LFE and SF, consistently perform well across model sizes and datasets, frequently ranking among the top two methods in our setup. ToMe also shows competitive performance, especially on the Small and Tiny models with the ESC50 dataset, but its inconsistency on the AST-B model may suggest sensitivity to model scale or dataset characteristics. We hypothesize that this may be due to ToMe's merging strategy, which can interfere with token semantics more noticeably in larger models where attention patterns are more specialized (Devlin et al., 2019).

Compute Cost Analysis. We report compute cost in terms of FLOPs for AS-20K using the AST-B model in Table 1. As expected, the number of FLOPs decreases consistently with lower keep rates across all datasets and methods, reflecting the expected reduction in attention computation as fewer tokens are retained. These trends validate that sparsification



(a) 示例梅尔谱图（Warden, 2018）。(b) 对应的幅度谱。

图5. 令牌级别的频谱平坦度。(a) 带有两个高亮令牌的声谱图。(b) 对应令牌嵌入的FFT显示令牌A具有音调峰值，而令牌B具有更平坦、噪声般的频谱——使得令牌A可能更具信息量。

4 实验设置

数据集。对于本次分析，选择了三个成熟的数据集，每个数据集涵盖不同的音频样本长度和任务复杂度。Speech Commands V2（SPC）（Warden, 2018）包含105,829条1秒钟的语音命令录音，覆盖35个不同的口令。该数据集被划分为84,843条训练样本、9,981条验证样本和11,005条测试样本。环境声音分类（ESC50）（Piczak, 2015）由2,000条5秒钟的环境声音录音组成，分为50个类别。其5折交叉验证设置每折包含1,600条训练样本，保证了可靠的评估框架。AudioSet（AS-20K）（Gemmeke 等, 2017）是一个大规模多类别数据集，包含来自YouTube的10秒音频片段，并标注为527个类别。平衡训练集约有22,000条样本，而评估集约有20,000条样本。

骨干网络。我们采用 AST（音频频谱变换器）(Gong 等，2021) 架构作为所有实验的骨干网络。稀疏化方法被整合到在 ImageNet (Deng 等，2009) 上预训练的 AST 模型中，使用基于标准 DeiT (Touvron 等，2020) 的配置：Base (B)、Small (S) 和 Tiny (T)，补丁大小为 16×16，时间和频率方向的重叠为 6，与默认的 AST 配方一致。

稀疏分配。与以往的工作一样，令牌稀疏模块被放置在AST每个变换器块的注意力和多层感知器（MLP）分支之间（Bolya 等，2023）。最常见且有效的做法是在变换器的三个等间隔点应用稀疏化，有效地分割-将网络分为四个部分（Kong 等, 2022；Liang 等, 2022b；Long 等, 2022）。这是我们实验中采用的配置。在每个稀疏化阶段，固定的保留率 $r \in \{0.25, 0.50, 0.70, 0.90\}$ 在所有稀疏化层中均匀应用。

实现细节。所有训练和微调过程均遵循原始 AST 设置（Gong 等，2021）。具体来说，AS-20K 和 ESC50 微调 25 个周期，而 SPC 训练 30 个周期。当启用标记稀疏化时，AST 中的所有默认掩码策略（如 SpecAugment（Park 等，2019））均被禁用以避免冗余，因为稀疏化本质上执行了一种数据掩码形式。所有模型变体（AST-B、AST-S、AST-T）均在训练后稀疏化设置下进行评估，以评估不同模型规模的性能和效率。对于稀疏微调实验，我们仅使用 AST-B，因为它是原始 AST 实现中的默认模型大小。为了量化不同稀疏化方法相关的计算成本，我们报告每个实例的浮点运算次数（FLOPs），使用批量大小为 1，通过 DeepSpeed 分析器（Aminabadi 等，2020）测量。

5 结果与分析

5.1 训练后稀疏化

不同方法和模型规模的性能。表 2a、2b 和 2c 分别报告了 AST-B、AST-S 和 AST-T 模型的性能。结果显示了所有数据集（SPC、ESC50、AS-20K）的情况，以及从 90% 到 25% 的令牌保留率。我们报告了三次运行的平均值。

从这些结果中，我们发现使用词元频率信息的方法，特别是 LFE 和 SF，在不同模型规模和数据集上表现稳定，经常在我们的设置中名列前两位。ToMe 也显示出具有竞争力的性能，特别是在使用 ESC50 数据集的小型和微型模型上，但它在 AST-B 模型上的表现不稳定可能表明其对模型规模或数据集特性敏感。我们假设这可能是由于 ToMe 的合并策略，在较大模型中，由于注意力模式更为专门化，这种策略会更明显地干扰词元语义（Devlin 等，2019）。

计算成本分析。我们在表1中使用AST-B模型报告了AS-20K的计算成本（以FLOPs为单位）。如预期的那样，在所有数据集和方法中，随着保留率的降低，FLOPs数量持续下降，这反映了随着保留的令牌减少，注意力计算的预期减少。这些趋势验证了稀疏化

Table 1. GFLOPs of AST-B model on AS-20K after sparsification across keep rates (r). ↓ denotes percentage reductions over baseline. Maximum reductions per keep rate are highlighted in green.

	AS-20K (GFLOPs)			
Baseline	261.43			
r (%)	90	70	50	25
Top-K (Haurum et al., 2022a)	222.28$^{\downarrow 15.0\%}$	164.16$^{\downarrow 37.2\%}$	124.85$^{\downarrow 52.3\%}$	93.74$^{\downarrow 64.3\%}$
EViT (Liang et al., 2022a)	222.61$^{\downarrow 14.8\%}$	164.45$^{\downarrow 37.1\%}$	125.05$^{\downarrow 52.2\%}$	93.99$^{\downarrow 64.0\%}$
ToMe (Bolya et al., 2023)	226.11$^{\downarrow 13.5\%}$	165.77$^{\downarrow 36.6\%}$	125.60$^{\downarrow 52.0\%}$	–
Intensity	222.28$^{\downarrow 15.0\%}$	164.16$^{\downarrow 37.2\%}$	124.89$^{\downarrow 52.2\%}$	93.79$^{\downarrow 64.1\%}$
LFE (Wang et al., 2022b)	222.66$^{\downarrow 14.8\%}$	164.46$^{\downarrow 37.1\%}$	125.10$^{\downarrow 52.1\%}$	93.95$^{\downarrow 64.1\%}$
SF	222.41$^{\downarrow 14.9\%}$	164.26$^{\downarrow 37.1\%}$	124.95$^{\downarrow 52.2\%}$	93.84$^{\downarrow 64.1\%}$

mechanisms function as intended, with lower keep rates leading to correspondingly lower compute cost. While this downward trend is consistent across methods, the actual savings differ slightly due to method-specific overheads. To illustrate the trade-off between performance and compute cost, we present Pareto front plots. In Fig. 1, we show results for AS-20K, while Fig. 6 shows SPC and ESC50 datasets, on AST-B. We find that frequency-domain methods such as **Spectral Flatness (SF)** and **Low-Frequency Energy (LFE)** consistently yield the highest performance across datasets, despite slightly higher computational overhead compared to simpler methods like Top-K. Results follow similar trends, showing the performance scalability of most methods across model sizes and tasks.

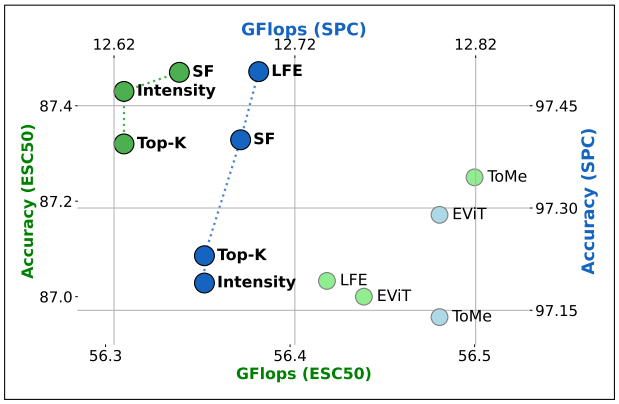


Figure 6. Post-training sparsification results on the AST-B (Gong et al., 2021) model evaluated on SPC and ESC50 at a 50% keep rate. Methods leveraging frequency-domain cues, such as **Spectral Flatness (SF)** and **Low-Frequency Energy (LFE) (Wang et al., 2022b)**, outperform other methods in terms of accuracy.

5.2 Sparsified Fine-tuning

Fig. 7 shows how task performance varies with decreasing keep rates across datasets under sparsified fine-tuning, with results averaged over three runs.

Across all datasets, we observe that performance remains robust even under aggressive sparsification, with most methods maintaining accuracy close to the baseline at a 25% keep rate. This indicates that when sparsification is integrated during training, the model is able to adapt to the imposed sparsity, learning to preserve task-relevant information despite reduced number of tokens. Furthermore, the performance differences between methods are less pronounced than in the post-training setting. This suggests that the model learns to accommodate whichever tokens are retained during training, thereby reducing sensitivity to the specific scoring heuristic. As a result, even simple methods like Top-K remain competitive, and the relative advantage of more informed strategies, such as SF and LFE, is reduced.

In summary, sparsified fine-tuning makes the model less sensitive to the specific sparsification heuristic, as it learns to adapt to the retained tokens. This contrasts with post-training sparsification, where token importance is fixed and the choice of heuristic plays a more decisive role.

5.3 Comparative & Controlled Studies

Contribution of Frequency-domain Information. To understand the source of performance differences between SF and LFE, we conduct a comparative study that isolates the effects of their frequency metrics and attention formulations. SF uses standard CLS-token attention (TopK-CLS), and LFE on the other hand, uses a head-weighted attention variant (TopK-Wt) (Wang et al., 2022b). To isolate the impact of these factors, we evaluate all four combinations of frequency metric (SF or LFE) and attention type (TopK-CLS or TopK-Wt). Each configuration is run five times to account for randomness.

As shown in Fig. 8, SF consistently outperforms its TopK-only counterparts across keep rates and attention variants. In contrast, while LFE generally improves over its baseline, it occasionally underperforms on AS-20K and ESC50, suggesting that emphasizing low-frequency components may not universally benefit performance and may depend on dataset characteristics. Furthermore, TopK-Wt tends to out-

表 1. AST-B 模型在 AS-20K 上经过稀疏化后的 GFLOPs，不同保留率下的结果（r）。↓ 表示相对于基线的百分比减少。每个保留率下的最大减少值已用绿色标出。

	AS-20K (GFLOPs)			
Baseline	261.43			
r (%)	90	70	50	25
Top-K (Haurum et al., 2022a)	$222.28^{\downarrow 15.0\%}$	$164.16^{\downarrow 37.2\%}$	$124.85^{\downarrow 52.3\%}$	$93.74^{\downarrow 64.3\%}$
EViT (Liang et al., 2022a)	$222.61^{\downarrow 14.8\%}$	$164.45^{\downarrow 37.1\%}$	$125.05^{\downarrow 52.2\%}$	$93.99^{\downarrow 64.0\%}$
ToMe (Bolya et al., 2023)	$226.11^{\downarrow 13.5\%}$	$165.77^{\downarrow 36.6\%}$	$125.60^{\downarrow 52.0\%}$	–
Intensity	$222.28^{\downarrow 15.0\%}$	$164.16^{\downarrow 37.2\%}$	$124.89^{\downarrow 52.2\%}$	$93.79^{\downarrow 64.1\%}$
LFE (Wang et al., 2022b)	$222.66^{\downarrow 14.8\%}$	$164.46^{\downarrow 37.1\%}$	$125.10^{\downarrow 52.1\%}$	$93.95^{\downarrow 64.1\%}$
SF	$222.41^{\downarrow 14.9\%}$	$164.26^{\downarrow 37.1\%}$	$124.95^{\downarrow 52.2\%}$	$93.84^{\downarrow 64.1\%}$

mechanisms function as intended, with lower keep rates leading to correspondingly lower compute cost. While this downward trend is consistent across methods, the actual savings differ slightly due to method-specific overheads. To illustrate the trade-off between performance and compute cost, we present Pareto front plots. In Fig. 1, we show results for AS-20K, while Fig. 6 shows SPC and ESC50 datasets, on AST-B. We find that frequency-domain methods such as Spectral Flatness (SF) and Low-Frequency Energy (LFE) consistently yield the highest performance across datasets, despite slightly higher computational overhead compared to simpler methods like Top-K. Results follow similar trends, showing the performance scalability of most methods across model sizes and tasks.

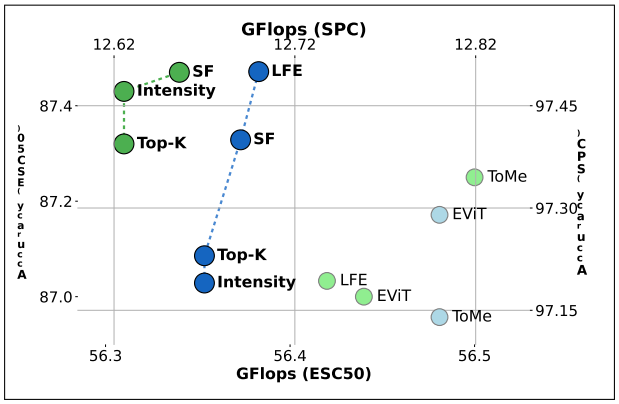


图6. 在AST-B（Gong等，2021）模型上进行训练后稀疏化的结果，在50%保留率下评估SPC和ESC50。利用频域线索的方法，如谱平坦度（SF）和低频能量（LFE）（Wang等，2022b），在准确率方面优于其他方法。

5.2 稀疏微调

图7显示了在稀疏化微调下，随着保留率降低，各数据集的任务性能变化，结果为三次运行的平均值。

在所有数据集上，我们观察到即使在激进的稀疏化下，性能仍然保持稳健，大多数方法在 25% 保留率下的准确率接近基线。这表明，当在训练过程中引入稀疏化时，模型能够适应施加的稀疏性，学习在减少的 token 数量下保留任务相关的信息。此外，各方法之间的性能差异比在训练后设置中要不明显。这表明模型在训练过程中学会适应保留下来的 token，从而减少对特定评分启发式的敏感性。因此，即使像 Top-K 这样简单的方法仍具有竞争力，而信息更多的策略，如 SF 和 LFE 的相对优势则降低。

总之，稀疏化微调使模型对特定的稀疏化启发式方法不太敏感，因为它学会了适应保留的标记。这与训练后稀疏化形成对比，后者标记的重要性是固定的，启发式方法的选择起着更决定性的作用。

5.3 比较与对照研究

频域信息的贡献。为了理解 SF 和 LFE 之间性能差异的来源，我们进行了一个比较研究，以隔离它们的频率度量和注意力机制的影响。SF 使用标准的 CLS-token 注意力（TopK-CLS），而 LFE 则使用一个头加权注意力变体（TopK-Wt）（Wang 等，2022b）。为了隔离这些因素的影响，我们评估了频率度量（SF 或 LFE）和注意力类型（TopK-CLS 或 TopK-Wt）的所有四种组合。每种配置运行五次以考虑随机性。

如图8所示，SF在各保持率和注意力变体下始终优于仅使用TopK的对应方法。相比之下，虽然LFE通常优于其基线，但在AS-20K和ESC50上偶尔表现不佳，这表明强调低频成分可能并不总能提高性能，并且可能依赖于数据集的特性。此外，TopK-Wt倾向于...

Table 2. Post-training sparsification results across model sizes.

(a) Post-training sparsification results for the **AST-B** model across keep rates (r). Best performing methods are in **bold**, second-best are underlined, and results outperforming the AST-B baseline are highlighted. SF consistently achieves the best performance across most keep rates and datasets, while LFE also performs competitively, particularly on SPC.

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-B	97.67				89.42				0.3572			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	97.46	97.43	97.21	88.73	89.02	88.42	87.32	78.47	0.3557	0.3546	0.3470	0.2956
EViT (Liang et al., 2022a)	97.44	97.27	97.29	90.03	88.77	88.17	87.00	75.42	0.3552	0.3545	0.3476	0.2981
ToMe (Bolya et al., 2023)	97.51	97.47	97.14	–	88.76	88.45	87.25	–	0.3567	0.3539	0.3470	–
Intensity	97.55	97.46	97.19	88.63	89.10	88.52	87.43	78.62	0.3567	0.3549	0.3479	**0.3033**
LFE (Wang et al., 2022b)	97.66	97.62	**97.50**	**93.49**	89.12	88.45	87.02	78.92	0.3568	0.3546	0.3484	0.3012
SF	**97.68**	**97.66**	97.40	89.00	**89.20**	**88.67**	**87.47**	**78.97**	**0.3576**	**0.3564**	**0.3499**	0.3007

(b) Post-training sparsification results for the **AST-S** model across keep rates (r). Best performing methods are in **bold**, second-best are underlined, and results outperforming the AST-S baseline are highlighted. SF and LFE dominate across datasets, mirroring the trends observed with AST-B. ToMe performs competitively on ESC50, indicating that token merging may be effective for smaller models.

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-S	97.21				83.75				0.3105			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	96.97	97.04	96.83	83.60	83.10	81.65	79.23	69.03	0.3058	0.3037	0.2995	0.2688
EViT (Liang et al., 2022a)	96.96	97.02	96.84	89.39	83.03	81.65	79.30	69.40	0.3052	0.3018	0.2950	0.2672
ToMe (Bolya et al., 2023)	97.16	96.93	96.32	–	**83.33**	**83.15**	**80.80**	–	0.3057	0.3029	0.3001	–
Intensity	96.89	96.95	96.76	83.33	83.08	82.58	78.03	**69.60**	0.3068	0.3043	0.2996	0.2734
LFE (Wang et al., 2022b)	**97.21**	97.24	**97.21**	**93.74**	83.05	81.50	78.88	68.33	0.3089	0.3068	0.3037	0.2792
SF	**97.21**	**97.25**	97.20	82.37	83.20	81.75	79.23	69.35	**0.3093**	**0.3074**	**0.3049**	**0.2799**

(c) Post-training sparsification results for the **AST-T** model across keep rates (r). Best performing methods are in **bold**, second-best are underlined, and results outperforming the AST-T baseline are highlighted. SF and LFE rank quite consistently among the top, showing strong scalability across model sizes, while ToMe excels on ESC50 but is less consistent across other datasets.

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-T	96.99				81.15				0.2759			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	96.37	96.21	95.93	88.06	80.58	78.65	73.95	56.23	0.2721	0.2699	0.2601	0.2189
EViT (Liang et al., 2022a)	96.35	96.19	95.87	**92.37**	80.35	78.53	73.78	**57.03**	0.2668	0.2632	0.2591	0.2161
ToMe (Bolya et al., 2023)	97.01	96.26	95.00	–	**81.25**	**79.53**	74.13	–	**0.2731**	0.2701	0.2603	–
Intensity	**97.02**	96.62	96.12	89.12	80.47	79.01	73.61	54.91	0.2695	0.2654	0.2599	0.2233
LFE (Wang et al., 2022b)	97.00	**97.03**	**96.81**	91.36	80.40	78.70	73.45	53.68	0.2699	0.2675	0.2624	0.2391
SF	**97.02**	96.98	96.71	87.87	80.63	78.70	**74.20**	56.73	0.2728	**0.2709**	**0.2638**	**0.2399**

perform TopK-CLS, especially at lower keep rates ($< 70\%$), which may be because TopK-Wt also includes peer-token attention, offering a more distributed importance signal than CLS-token attention alone.

Overall, incorporating frequency-domain information yields modest but consistent gains over attention-only pruning methods in post-training settings.

Effect of Input Duration on Intensity-Based Pruning. In Section 5.1 (Table 2a), we observed that intensity-based pruning outperforms purely attention-based methods (Top-K, EViT) on ESC50 and AS-20K across most settings, whereas its advantage is less pronounced on SPC. This relatively strong performance on datasets with longer audio clips (5 and 10 seconds, respectively) - compared to weaker results on SPC (with 1-second clips) raises the question of a potential dependency on input duration.

To investigate this, we conduct a controlled experiment comparing intensity-based pruning with a purely attention-based Top-K method. We apply random temporal windowing at inference time on AS-20K and ESC50, simulating input

表2. 不同模型规模的训练后稀疏化结果。(a) AST-B模型在不同保留率(r)下的训练后稀疏化结果。表现最佳的方法用粗体表示，次优方法用下划线表示，超过AST-B基线的结果已突出显示。SF在大多数保留率和数据集上始终表现最佳，而LFE的表现也具有竞争力，特别是在SPC上。

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-B	97.67				89.42				0.3572			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	97.46	97.43	97.21	88.73	89.02	88.42	87.32	78.47	0.3557	0.3546	0.3470	0.2956
EViT (Liang et al., 2022a)	97.44	97.27	97.29	90.03	88.77	88.17	87.00	75.42	0.3552	0.3545	0.3476	0.2981
ToMe (Bolya et al., 2023)	97.51	97.47	97.14	–	88.76	88.45	87.25	–	0.3567	0.3539	0.3470	–
Intensity	97.55	97.46	97.19	88.63	89.10	88.52	87.43	78.62	0.3567	0.3549	0.3479	**0.3033**
LFE (Wang et al., 2022b)	97.66	97.62	**97.50**	**93.49**	89.12	88.45	87.02	78.92	0.3568	0.3546	0.3484	0.3012
SF	**97.68**	**97.66**	97.40	89.00	**89.20**	**88.67**	**87.47**	**78.97**	**0.3576**	**0.3564**	**0.3499**	0.3007

(b) AST-S 模型在不同保留率 (r) 下的训练后稀疏化结果。表现最好的方法以粗体显示，次优方法以下划线显示，超过 AST-S 基线的结果已突出显示。SF 和 LFE 在各数据集上占主导地位，这与 AST-B 所观察到的趋势相似。ToMe 在 ESC50 上表现具有竞争力，这表明令牌合并可能对较小的模型有效。

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-S	97.21				83.75				0.3105			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	96.97	97.04	96.83	83.60	83.10	81.65	79.23	69.03	0.3058	0.3037	0.2995	0.2688
EViT (Liang et al., 2022a)	96.96	97.02	96.84	89.39	83.03	81.65	79.30	69.40	0.3052	0.3018	0.2950	0.2672
ToMe (Bolya et al., 2023)	97.16	96.93	96.32	–	**83.33**	**83.15**	**80.80**	–	0.3057	0.3029	0.3001	–
Intensity	96.89	96.95	96.76	83.33	83.08	82.58	78.03	**69.60**	0.3068	0.3043	0.2996	0.2734
LFE (Wang et al., 2022b)	**97.21**	97.24	**97.21**	**93.74**	83.05	81.50	78.88	68.33	0.3089	0.3068	0.3037	0.2792
SF	**97.21**	**97.25**	97.20	82.37	83.20	81.75	79.23	69.35	**0.3093**	**0.3074**	**0.3049**	**0.2799**

(AST-T 模型在不同保留率（r）下的后训练稀疏化结果。表现最好的方法用粗体表示，次优方法用下划线表示，超过 AST-T 基线的结果已高亮显示。SF 和 LFE 在排名上相当稳定，显示出对模型规模的强大可扩展性，而 ToMe 在 ESC50 上表现优异，但在其他数据集上的表现不太一致。

	SPC (Acc(%))				ESC50 (Acc(%))				AS-20K (mAP)			
AST-T	96.99				81.15				0.2759			
***r* (%)**	90	70	50	25	90	70	50	25	90	70	50	25
Top-K (Haurum et al., 2022a)	96.37	96.21	95.93	88.06	80.58	78.65	73.95	56.23	0.2721	0.2699	0.2601	0.2189
EViT (Liang et al., 2022a)	96.35	96.19	95.87	**92.37**	80.35	78.53	73.78	**57.03**	0.2668	0.2632	0.2591	0.2161
ToMe (Bolya et al., 2023)	97.01	96.26	95.00	–	**81.25**	**79.53**	74.13	–	**0.2731**	0.2701	0.2603	–
Intensity	**97.02**	96.62	96.12	89.12	80.47	79.01	73.61	54.91	0.2695	0.2654	0.2599	0.2233
LFE (Wang et al., 2022b)	97.00	**97.03**	**96.81**	91.36	80.40	78.70	73.45	53.68	0.2699	0.2675	0.2624	0.2391
SF	**97.02**	96.98	96.71	87.87	80.63	78.70	**74.20**	56.73	0.2728	**0.2709**	**0.2638**	**0.2399**

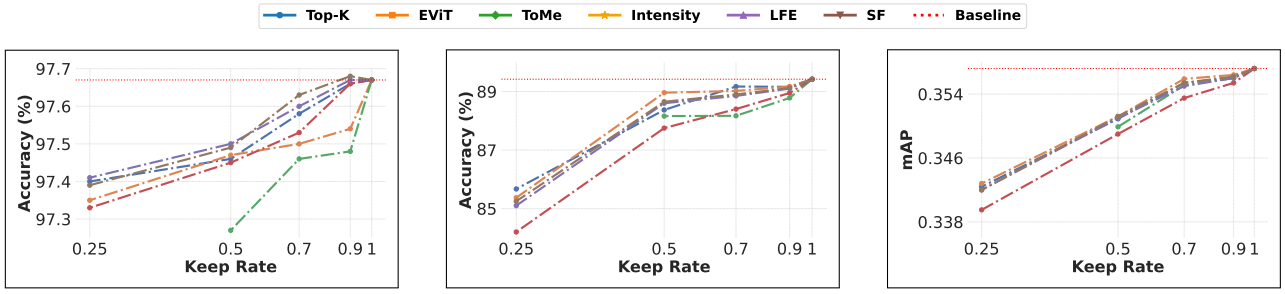
执行 TopK-CLS，特别是在较低保留率下（< 70%），这可能是因为 TopK-Wt 也包括同伴 token 注意力，提供比仅 CLS token 注意力更分散的重要性信号。

总体而言，在训练后设置中，结合频域信息相比仅使用注意力的剪枝方法能带来适度但稳定的提升。

输入持续时间对基于强度剪枝的影响。在第5.1节（表2a）中，我们观察到基于强度的

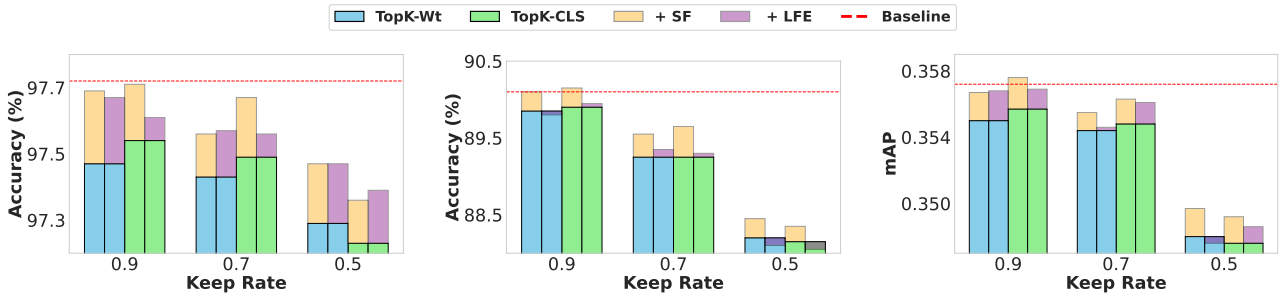
在ESC50和AS-20K的大多数设置中，剪枝的表现优于纯基于注意力的方法（Top-K, EViT），而在SPC上其优势则不那么明显。在较长音频片段（分别为5秒和10秒）的数据集上表现相对较强——相比之下，在SPC（1秒片段）上的结果较弱——这提出了一个关于可能依赖于输入时长的问题。

为了调查这一点，我们进行了一个受控实验，将基于强度的剪枝与纯注意力的Top-K方法进行比较。在推理阶段，我们在AS-20K和ESC50上应用随机时间窗口，模拟输入



(a) SPC (b) ESC50 (c) AS-20K

Figure 7. **Sparsified Fine-tuning:** Performance across varying keep rates. Frequency-based methods (SF, LFE), as well as pure attention-based pruning methods (Top-K, EViT), consistently maintain high performance across tasks. Most methods retain competitive performance down to a 50% keep rate. Overall, the performance differences between methods are not very significant, indicating that multiple sparsification methods can be equally effective when fine-tuned.



(a) SPC (b) ESC50 (c) AS-20K

Figure 8. Performance comparison of frequency-based methods (LFE and SF) against attention-only baselines (TopK-Wt and TopK-CLS). SF slightly but consistently outperforms attention-based baselines across keep rates, while LFE shows competitive results but occasionally underperforms on ESC50 and AS-20K.

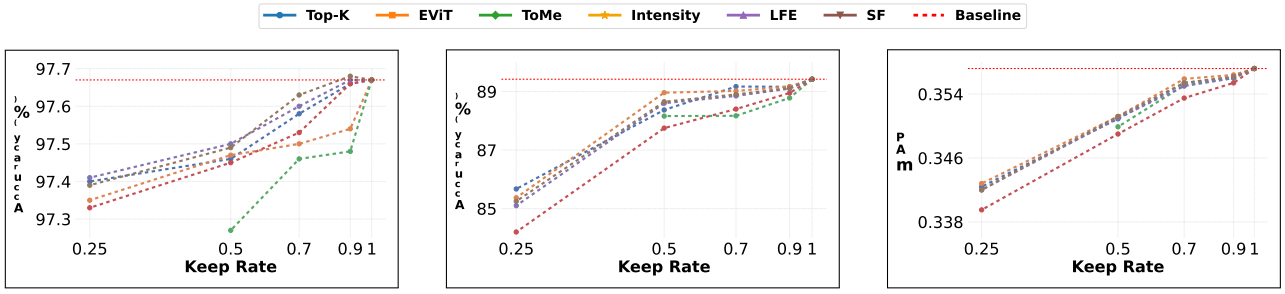
lengths ranging from 1 second to the full duration of each dataset (i.e., 1s to 10s for AS-20K, and 1s to 5s for ESC50). Each result is averaged over six runs to account for randomness. The sparsification configuration is fixed to the most aggressive setting, with a keep rate of only 25%.

As shown in Fig. 9, intensity-based pruning consistently outperforms the Top-K method for audio segments longer than 2 seconds across both datasets. These results provide further empirical support for our hypothesis and suggest that intensity-based pruning works better for longer inputs. We attribute this effect to the interaction between input duration and token redundancy. First, longer audio clips yield more tokens and often contain repeated or sustained events, introducing redundancy. Therefore, even after sparsification, enough informative content remains. Second, they tend to include clear low-intensity regions, such as silence or background noise, which can be safely discarded. In contrast, short clips (1-2 seconds) typically contain a single spoken word or sound event, where nearly all tokens contribute to the core signal. These inputs offer little redundancy and minimal variation in intensity, making it harder to distinguish unimportant tokens. As a result, sparsification based on intensity can mistakenly remove critical information, leading to performance degradation.

5.4 Practical Guidelines

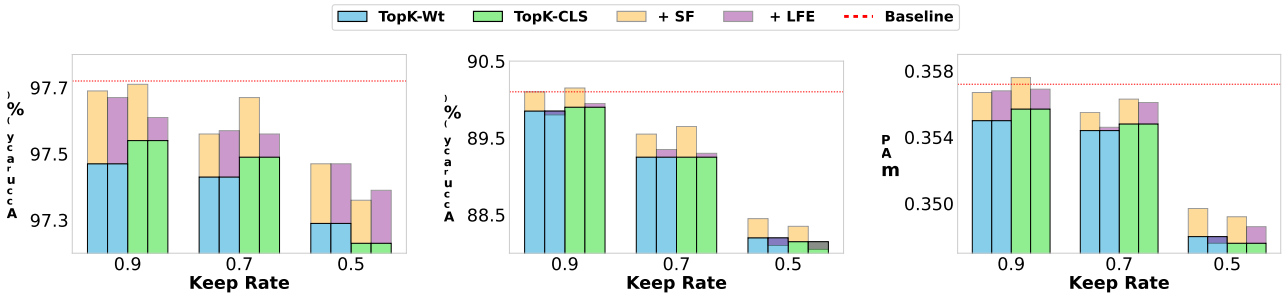
Based on our analysis, we outline several recommendations for effectively applying token sparsification in audio transformers:

- Frequency-aware heuristics, particularly Spectral Flatness (SF) and Low-Frequency Energy (LFE), quite consistently outperform other methods. SF stands out as one of the most practical choices for post-training sparsification, as it requires no additional tuning and generalizes well across configurations. LFE, while also effective, involves tuning a low-pass filter cutoff to maximize performance, which may require dataset-specific calibration.
- For scenarios involving longer input durations (e.g., $>$ 2 seconds), intensity-based pruning can also outperform other standard methods mainly for larger model sizes, but it requires threshold tuning to distinguish between high- and low-energy patches.



(a) SPC (b) ESC50 (c) AS-20K

图7. 稀疏微调：在不同保留率下的性能表现。基于频率的方法（SF, LFE）以及纯基于注意力的剪枝方法（Top-K, EViT）在各项任务中均能持续保持高性能。大多数方法在保留率降至50%时仍能保持竞争力的性能。总体来看，各方法之间的性能差异并不显著，这表明在微调时，多个稀疏化方法都可以同样有效。



(a) SPC (b) ESC50 (c) AS-20K

图 8. 基于频率的方法（LFE 和 SF）与仅使用注意力的基线（TopK-Wt 和 TopK-CLS）的性能比较。SF 在各保持率下略微但持续地优于基于注意力的基线，而 LFE 显示出竞争力的结果，但在 ESC50 和 AS-20K 上偶尔表现不佳。

长度从1秒到每个数据集的完整持续时间不等（即AS-20K为1秒到10秒，ESC50为1秒到5秒）。每个结果都是对六次运行的平均，以考虑随机性。稀疏化配置固定为最激进的设置，保留率仅为25%。

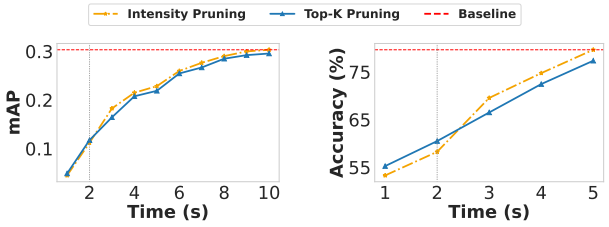
如图9所示，对于两个数据集中长度超过2秒的音频片段，基于强度的剪枝方法始终优于Top-K方法。这些结果进一步为我们的假设提供了实证支持，并表明基于强度的剪枝在处理较长输入时效果更佳。我们将这种效果归因于输入持续时间与标记冗余之间的相互作用。首先，较长的音频片段会产生更多的标记，并且通常包含重复或持续的事件，从而引入冗余。因此，即使在稀疏化之后，也会保留足够的有用内容。其次，它们往往包括清晰的低强度区域，如静音或背景噪声，这些部分可以被安全地舍弃。相反，较短的片段（1-2秒）通常只包含一个单词或声音事件，其中几乎所有标记都对核心信号有贡献。这些输入提供的冗余很少，并且强度变化最小，使得区分不重要的标记变得更加困难。因此，稀疏化紧张性可能错误地移除关键信息，从而导致性能下降。

5.4 实用指南

根据我们的分析，我们概述了在音频转换器中有效应用令牌稀疏化的几个建议：

- 频率感知的启发式方法，特别是谱平坦度（SF）和低频能量（LFE），通常比其他方法表现得更好。SF突显为后训练稀疏化中最实用的选择之一，因为它无需额外调优，并且在各种配置中具有良好的泛化能力。LFE虽然也有效，但需要调节低通滤波器的截止频率以最大化性能，这可能需要针对具体数据集进行校准。

- 对于涉及较长输入时长的场景（例如，>2 秒），基于强度的剪枝在较大模型尺寸下通常也能优于其他标准方法，但它需要阈值调节以区分高能量和低能量的片段。



(a) AS-20K (b) ESC50

Figure 9. Token pruning performance as a function of input duration. Intensity Pruning consistently outperforms Top-K pruning for longer audio windows (> 2 seconds), suggesting that spectrogram intensity is a more informative proxy for longer audio samples.

- While token merging methods like ToMe can perform well on small models or specific tasks, their effectiveness is more variable and less consistent across certain tasks.
- Finally, when applying sparsification during fine-tuning, the specific heuristic becomes less critical, as the model can adapt to sparsity - enabling even simple methods to perform well.

6 Conclusion

In this work, we presented a comprehensive evaluation of token sparsification techniques for audio transformers, covering pruning and merging approaches across multiple model sizes, datasets, and deployment scenarios. Our results show that frequency-aware methods, particularly those drawing on spectral flatness and low-frequency energy, consistently outperform attention-only methods, particularly in post-training sparsification settings. We also introduced a novel strategy that combines attention with spectral priors, which proved effective across a wide range of configurations. These findings underscore the value of integrating audio-specific cues into sparsification strategies, moving beyond generic attention signals. Looking forward, we see opportunities to explore more advanced task-specific token estimation strategies and extend these techniques to real-time applications.

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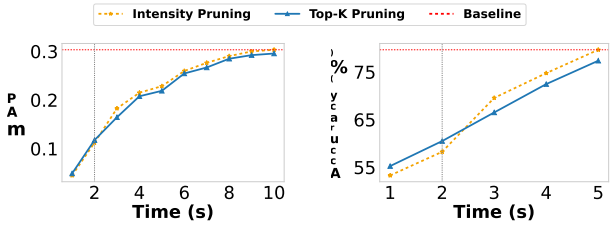
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(a) AS-20K (b) ESC50

图9. 令牌剪枝性能随输入时长的变化。对于较长的音频窗口（大于2秒），强度剪枝始终优于Top-K剪枝，这表明频谱图强度对于较长音频样本是更具信息量的代理。

- 虽然像 ToMe 这样的令牌合并方法在小模型或特定任务上表现良好，但它们的有效性在某些任务上的表现更为不稳定，且一致性较差。

- 最后，在微调过程中应用稀疏化时，具体的启发式方法变得不那么关键，因为模型可以适应稀疏性——即使是简单的方法也能表现良好。

6 结论

在这项工作中，我们对音频变压器的令牌稀疏化技术进行了全面评估，涵盖了不同模型大小、数据集和部署场景下的剪枝和合并方法。我们的结果显示，频率感知的方法，尤其是利用谱平坦度和低频能量的方法，在后训练稀疏化设置中持续优于仅基于注意力的方法。我们还引入了一种将注意力与谱先验相结合的新策略，在各种配置下都证明是有效的。这些发现强调了在稀疏化策略中整合音频特定线索的价值，超越了通用的注意力信号。展望未来，我们看到有机会探索更高级的特定任务令牌估计策略，并将这些技术扩展到实时应用中。

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